

Amrutha Thalla

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Academic Profile

MS graduate in Information Systems with a 3.7/4.0 GPA and experience developing RAG applications, evaluating machine learning models, and implementing LoRA fine-tuning. Seeking supervised research opportunities in Human-Centred Generative AI for Healthcare, with an emphasis on how people understand, verify, and appropriately rely on AI-generated health information.

Research Interests

Human-AI interaction in healthcare; evidence-grounded health information; LLM and RAG evaluation; explainable and trustworthy AI; visual explanations and patient understanding. Additional interest in multimodal generative AI.

Education

Master of Science in Information Systems

Indiana Institute of Technology, Fort Wayne, Indiana, USA | Aug 2023 – May 2025

GPA: 3.7/4.0 | Course-based degree

Study focus: Data Management, Cloud Computing, and System Architecture

Bachelor of Technology in Computer Science and Engineering

JNTUH, India | 2018

Selected Technical Projects

RAG Question Answering over Research Papers

- Built a question-answering pipeline over six foundational RAG papers using sentence-transformer embeddings and ChromaDB retrieval.
- Configured chunking, overlap, and top-k retrieval; constrained generation to retrieved context and displayed source citations in a Streamlit interface.
- Designed the interface to support checking generated answers against the retrieved source material.

Python, LangChain, sentence-transformers, ChromaDB, Groq API with Llama 3.1, Streamlit

LoRA Fine Tuning for Structured Answers

- Fine-tuned distilgpt2 for a fixed answer format using LoRA, training approximately 295,000 parameters, or 0.36% of the 82-million-parameter base model.
- Applied label masking to train on completion tokens; compared outputs with an untouched base model, including questions not used in training, and documented limitations.

Python, Hugging Face Transformers, PEFT, Google Colab

Selected Technical Projects Continued

Healthcare Insurance Cost Analysis and Prediction

- Analysed more than 1,300 insurance records using Python and SQL; observed that smokers had approximately three to four times higher average charges in the analysed dataset.
- Created BMI category, age band, and risk-flag features; compared regression approaches including Linear Regression, Random Forest, and XGBoost.
- Built Power BI and Excel dashboards for non-technical users and deployed a Streamlit interface for insurance-charge estimation.

Python, Pandas, Scikit-learn, XGBoost, SQL, Power BI, Excel, Streamlit

CrewAI and phidata Comparison

- Implemented the same two-agent research and summarisation task in both frameworks and presented outputs side by side in Streamlit.
- Observed that search-tool differences affected output quality on the selected recent topic; documented this limitation when interpreting the framework comparison.

Python, CrewAI, phidata, Gemini API, Serper, DuckDuckGo Search, Streamlit

Professional Experience

Software Engineer | Cortracker Inc

Feb 2025 – Nov 2025

- Developed SQL queries and stored procedures supporting reporting and data retrieval for HR and CRM systems.
- Built dashboards to communicate business KPIs and maintained REST APIs connecting front-end applications with back-end databases.
- Worked with product and operations teams to translate reporting and data requirements into software solutions.

SQL Server, ASP.NET Core, C#, Angular, REST APIs, Azure DevOps, Git

Technical Training

Full Stack Data Science and Gen AI Trainee | Naresh i Technologies

Dec 2025 – Present | Training programme in progress

- Practise data cleaning, exploratory analysis, classification, regression, and model comparison using Python and machine learning libraries.
- Develop document Q&A and chatbot applications using LLM APIs, LangChain, and RAG; create Streamlit interfaces and apply NLP preprocessing, embeddings, and semantic search.

Technical Skills

Programming: Python, SQL, C#, .NET Core

GenAI and NLP: RAG, LangChain, LangGraph, CrewAI, phidata, Hugging Face Transformers, PEFT and LoRA, OpenAI API, Gemini API, Ollama, NLTK, spaCy

Data and modelling: Pandas, NumPy, Scikit-learn, XGBoost, LightGBM, TensorFlow, Keras

Interfaces and visualisation: Streamlit, Flask, FastAPI, Power BI, Tableau, Matplotlib, Seaborn, HTML, CSS, JavaScript

Databases and deployment: SQL Server, MySQL, PostgreSQL, Azure, Docker, Git, CI/CD, REST APIs